QUOTIENTS OF REGULAR RINGS BY REGULAR SEQUENCES ARE COMPLETE INTERSECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Completion exactness. algebra-lemma-completion-tensor (cross-chapter, named). Identifies the completed quotient A^^ = R_p^^/(f) R_p^^.
- Regularity of completed local rings. more-algebra-lemma-completion-regular (cross-chapter, named). Certifies R_p^^ is a regular local ring, the base of the CI definition.

Source reference. Original reference 00MB: focused-seeds/geometry/S9-dpa-quotient-regular-ring-by-regular-sequence/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MA: focused-seeds/geometry/S9-dpa-quotient-regular-ring-by-regular-sequence/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00LM: focused-seeds/geometry/S9-dpa-quotient-regular-ring-by-regular-sequence/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07NY: focused-seeds/geometry/S9-dpa-quotient-regular-ring-by-regular-sequence/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a regular ring. Let $\mathfrak{p} \subset R$ be a prime. Let $f_1, \ldots, f_r \in \mathfrak{p}$ be a regular sequence. Then the completion of*

$$A = (R/(f_1, \ldots, f_r))_{\mathfrak{p}} = R_{\mathfrak{p}}/(f_1, \ldots, f_r)R_{\mathfrak{p}}$$

is a complete intersection in the sense defined above.

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Proof. The completion of A is equal to $A^\wedge = R^\wedge_\mathfrak{p}/(f_1, \ldots, f_r)R^\wedge_\mathfrak{p}$ because completion for finite modules over the Noetherian ring $R_\mathfrak{p}$ is exact (Algebra, Lemma 00MA). The image of the sequence $f_1, \ldots, f_r$ in $R_\mathfrak{p}$ is a regular sequence by Algebra, Lemmas 00MB and 00LM. Moreover, $R^\wedge_\mathfrak{p}$ is a regular local ring by More on Algebra, Lemma 07NY. Hence the result holds by our definition of complete intersection for complete local rings. $\square$